

25-12-05_update

Default Clustering behavior

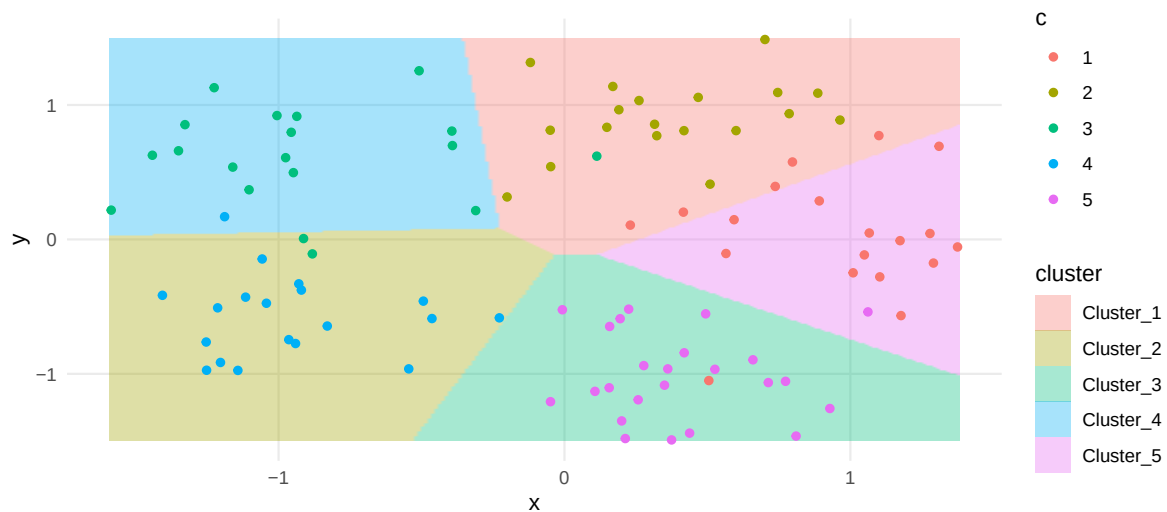
The available k means engines available with tidyclust require picking some number of clusters so the previous default was a rather arbitrary $k = 3$ k means model if the user didn't supply a method.

To get a stronger default the new behavior if no method is supplied is 5 fold cross validation across $k = 2$ to 10, where the k with the highest average silhouette value is selected. The user is warned this was selected and that value is used to plot.

This does require tidymodels to use.

```
samp <- rnorm_mixture(100, mean = roots_of_unity(5), var = .1)
df <- data.frame(x = samp[,1], y = samp[,2], c = attr(samp, "component"))
ggplot(df, aes(x, y))+
  geom_cluster()+
  geom_point(aes(col = c))
```

! a k means model with 5 clusters was used



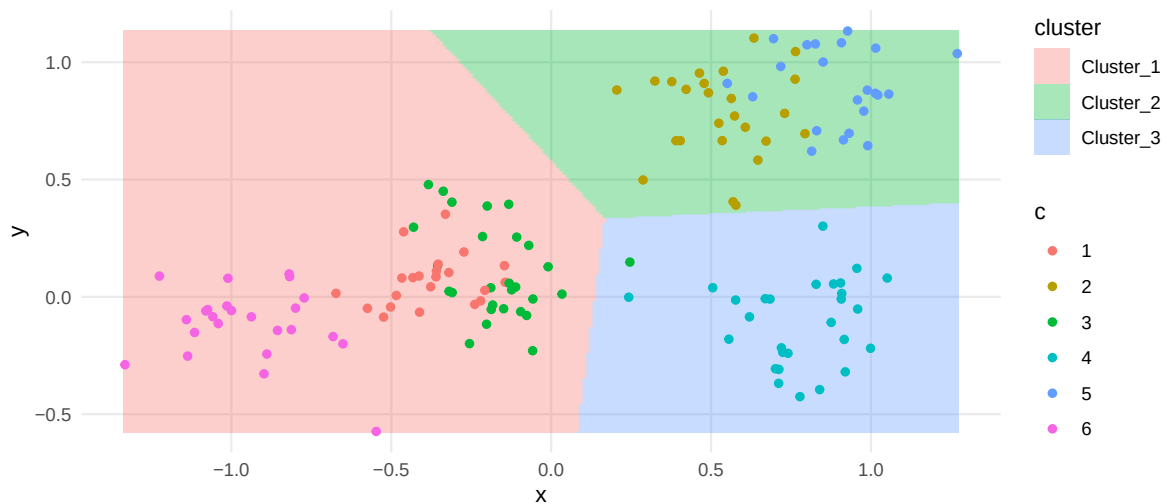
The model is pretty effective with Roots of Unity

```

set.seed(123)
means <- cbind(x = runif(6,-1), y = runif(6,-1))
samp <- rnorm_mixture(149, mean = means, var = .025)
bidata <- tibble(x = samp[,1], y = samp[,2], c = attr(samp, "component"))
ggplot(bidata, aes(x, y))+
  geom_cluster()+
  geom_point(aes(col = c))

```

! a k means model with 3 clusters was used



More complex data can be difficult though.

Documentation work

Man pages

I have condensed the documentation from `geom_classify`, `geom_classify_boundary`, and `geom_point_classify` into one page. as well as `geom_cluster`, `geom_cluster_boundary`, and `geom_point_cluster` into one page. This should help reduce the amount of copy paste documentation. I've further outlined the computed stats available to each geom.